



CPSC 100

Computational Thinking

Data Visualization

Instructor: Firas Moosvi
Department of Computer Science
University of British Columbia



Agenda

- Course Admin
 - Reminder about Milestone 4
- Gestalt Principles (Part 2)

Course Admin

Learning Goals



Learning Goals

After this **today's lecture**, you should be able to:

- Understand, define, name Gestalt Principles and explain their role in visualization
 - Demonstrate how these principles appear in real life—such as recognizing how proximity is used in store displays or how logos use closure to create memorable designs.
- Explore how understanding these principles can inform better visual communication, whether for creating infographics, arranging elements on a webpage, or organizing a physical space.

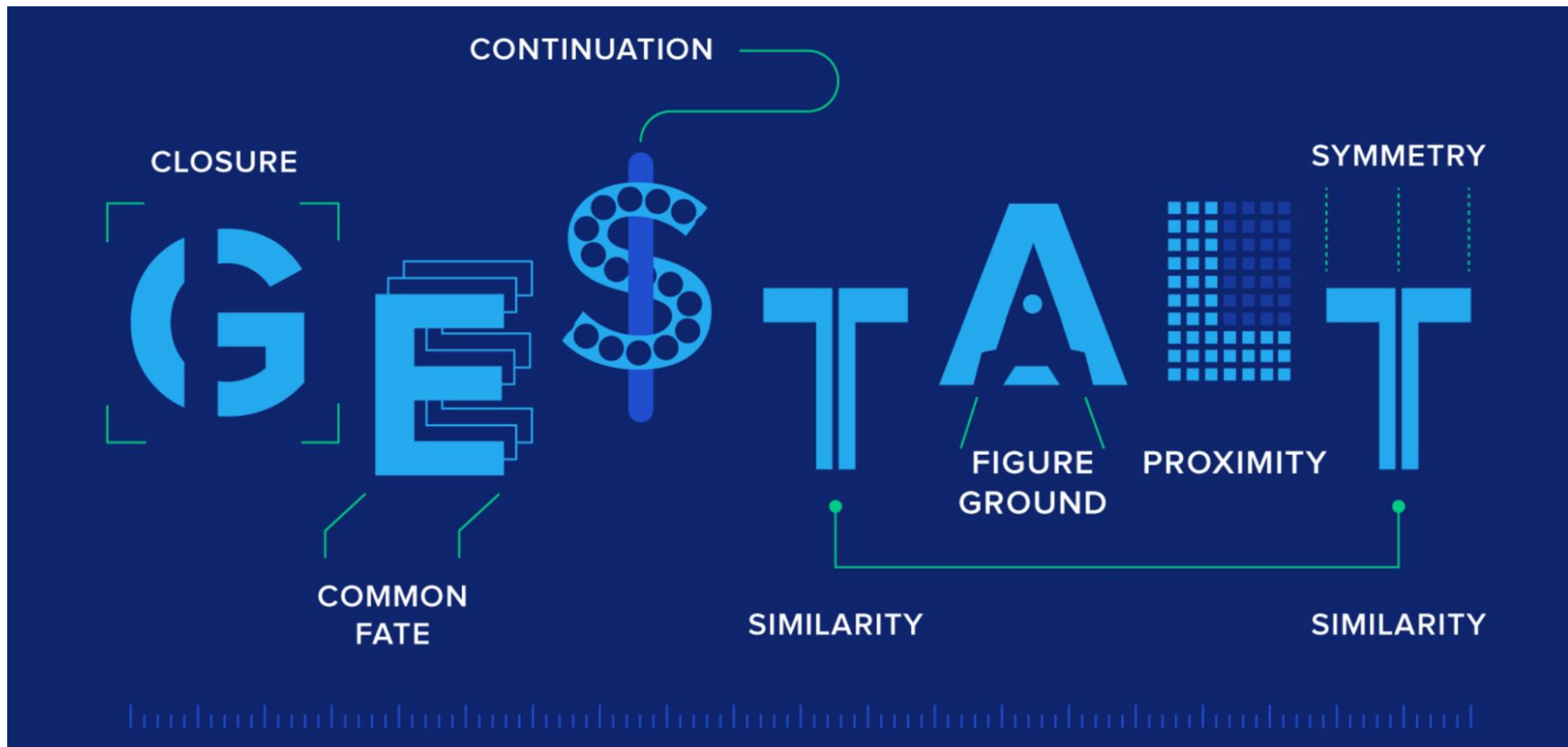
Gestalt Principles



What are Gestalt Principles?

- "Gestalt" is German for "unified whole".
- German psychologists Max Wertheimer, Kurt Koffka, and Wolfgang Kohler created the **Gestalt Principles** in the 1920s.
- They wanted to understand **how people make sense of the confusing things they see and hear**. They identified a set of laws that address the natural compulsion to find order in disorder. According to this, the mind "informs" what the eye sees by perceiving a series of individual elements as a whole.

What are Gestalt Principles?





Gestalt Principles

1. Emergence
2. Closure (Reification)
3. Common Region
4. Continuity or Continuation
5. Proximity
6. Multistability
7. Figure/Ground
8. Invariance
9. Pragnanz
10. Similarity
11. Symmetry and Order
12. Common Fate



Q: Which Gestalt Principles is most applicable to this logo?



- A. Emergence
- B. Closure
- C. Common Region
- D. Continuity
- E. Multistability





Q: Which Gestalt Principles is most applicable to this logo?



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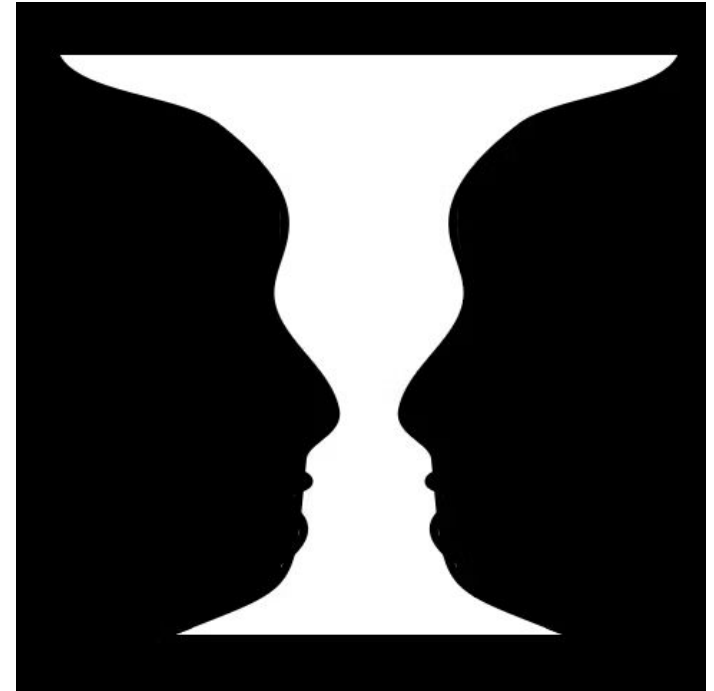




Q: Which Gestalt Principles is most applicable to this image?



iClicker



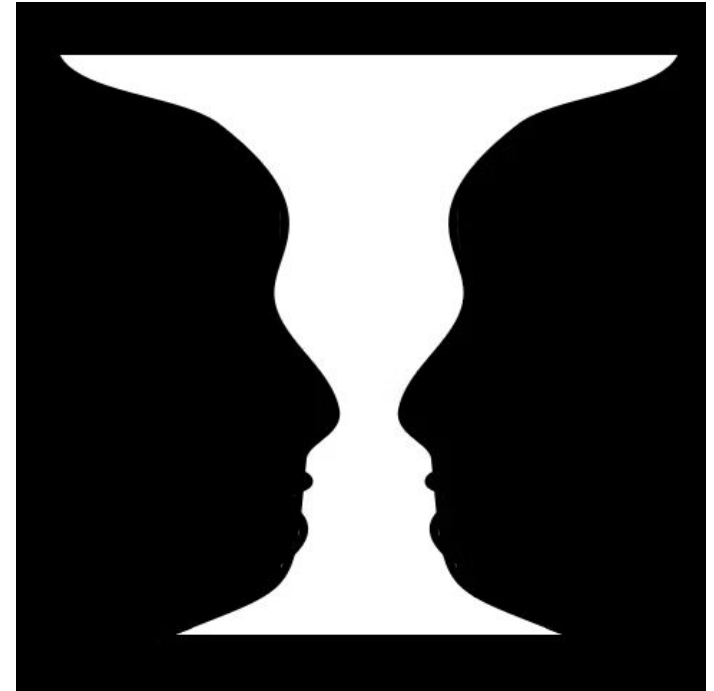
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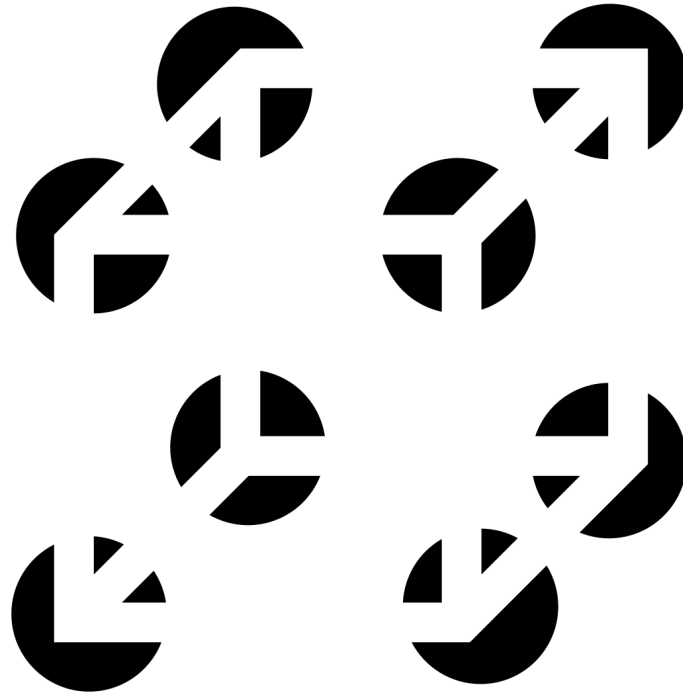
- A. Emergence
- B. Closure
- C. Common Region
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Review

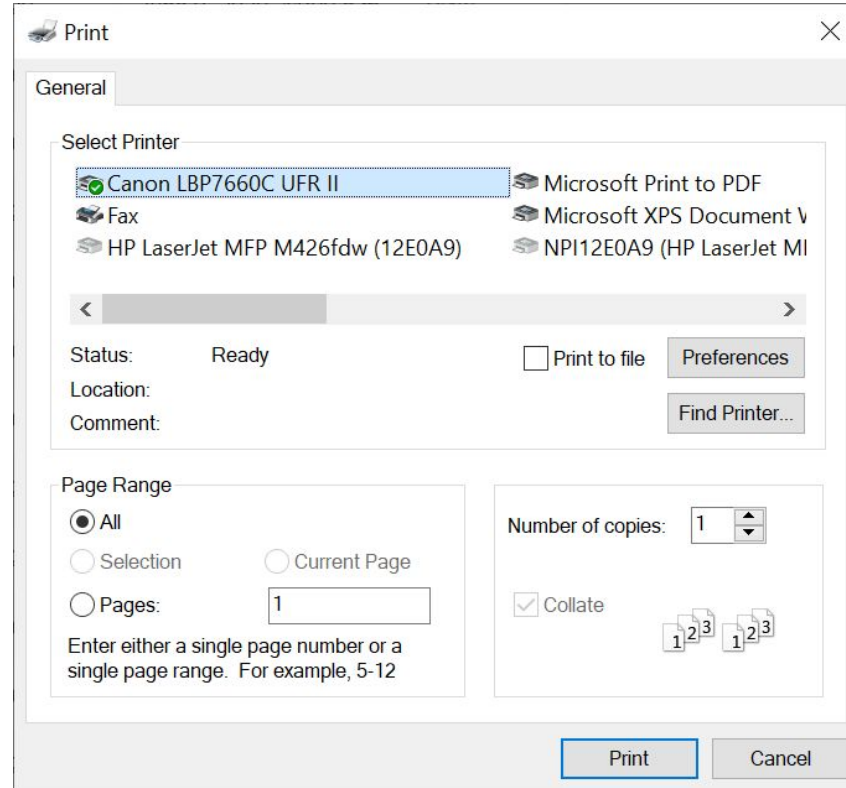
Gestalt Rule: **Emergence**



Gestalt Rule: **Closure**



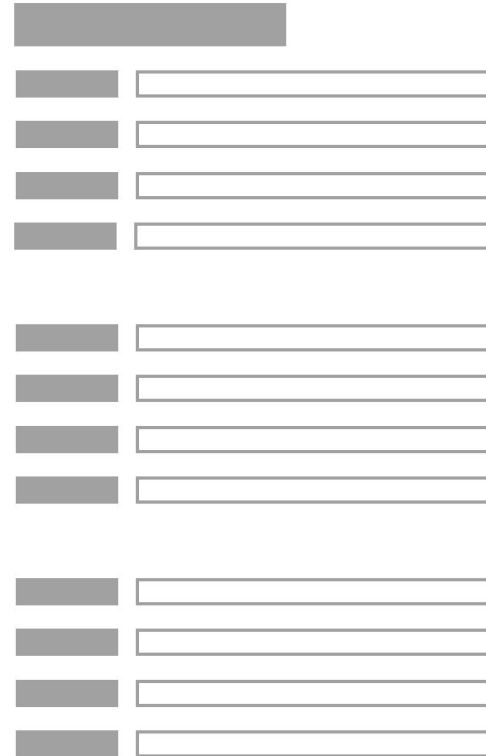
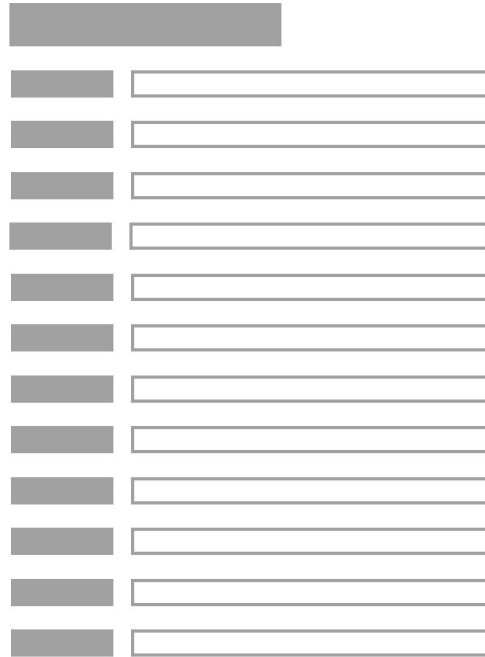
Gestalt Rule: **Common Region**



Gestalt Rule: **Continuity**



Gestalt Rule: **Proximity**



Gestalt Rule: **Multistability**





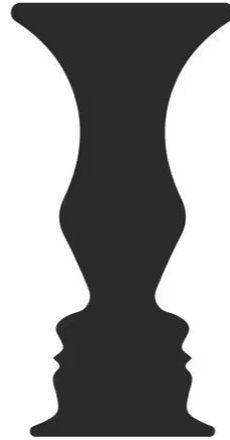


Gestalt Principles (Part 2)

- | | |
|-------------------------------|------------------------|
| 1. Emergence | 7. Figure/Ground |
| 2. Closure (Reification) | 8. Invariance |
| 3. Common Region | 9. Pragnanz |
| 4. Continuity or Continuation | 10. Similarity |
| 5. Proximity | 11. Symmetry and Order |
| 6. Multistability | 12. Common Fate |

Figure/Ground

Gestalt Rule: **Figure/Ground**



Rubin's Vase

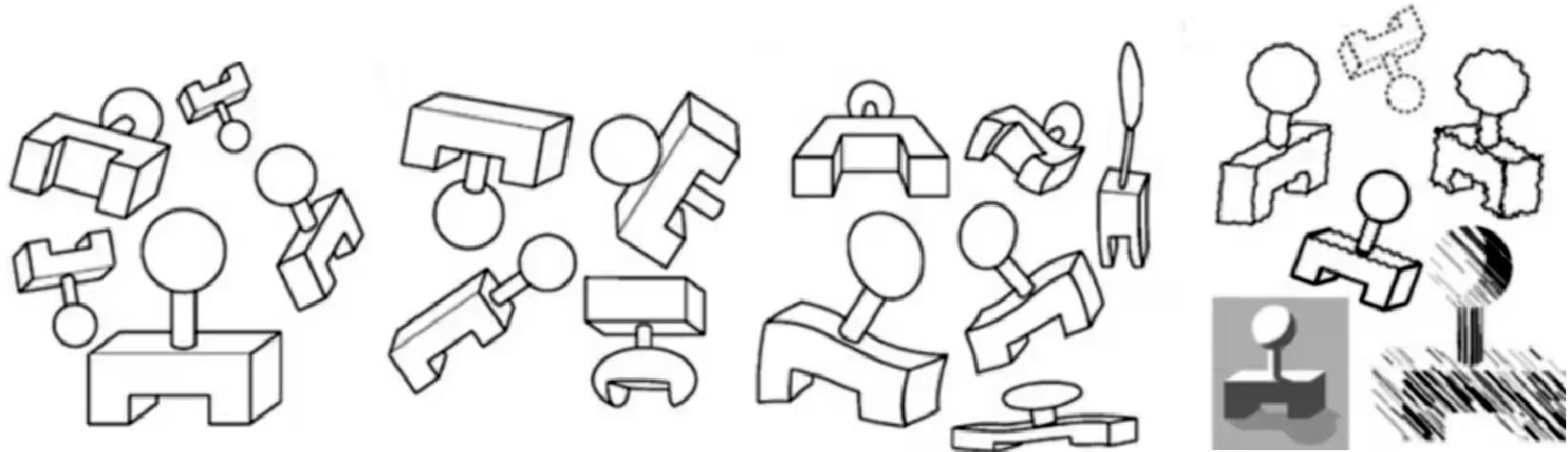


Figure/Ground

- We naturally separate visual elements into a “**figure**” (the main object) and a “**ground**” (the background). **Which element is figure can flip**, as in the vase-faces illusion.
 - Vital for understanding **how we pick out objects in a busy environment** (e.g., spotting a friend in a crowd).
 - Used in designs to create emphasis or visual interest. (e.g. infographics)
- E.g. When you read text on a page, the words are the figure and the white page is the ground. You focus on the figure (text) and let the ground (white background) fade away.

Invariance

Gestalt Rule: **Invariance**



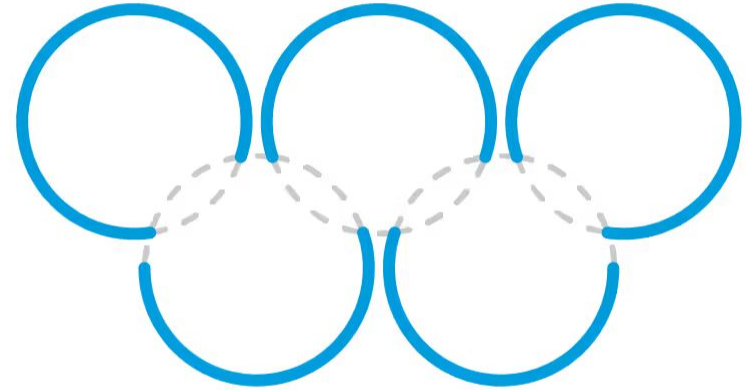
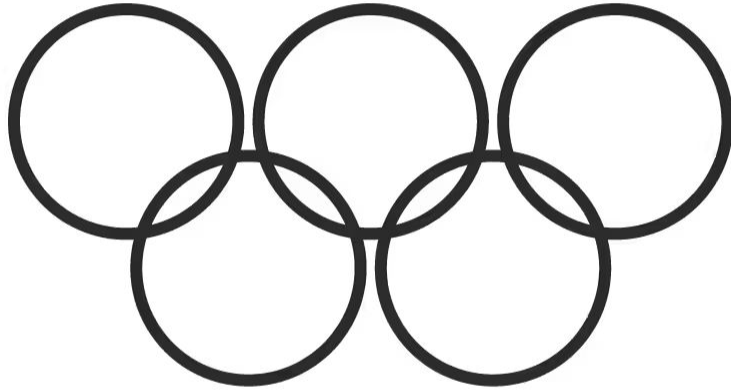


Invariance

- We can **recognize simple shapes regardless of rotation, scale, or slight distortions**. A circle is still seen as a circle even if it's tilted or partially covered.
 - Explains how we recognize letters in *different fonts* or everyday objects from various angles.
 - Helps us see objects as the “same” despite minor variations.
- E.g. Whether you see a coffee cup right-side up, upside-down, or at an angle, you still know it's a coffee cup. The shape and handle are recognizable enough.

Pragnanz

Gestalt Rule: **Pragnanz**

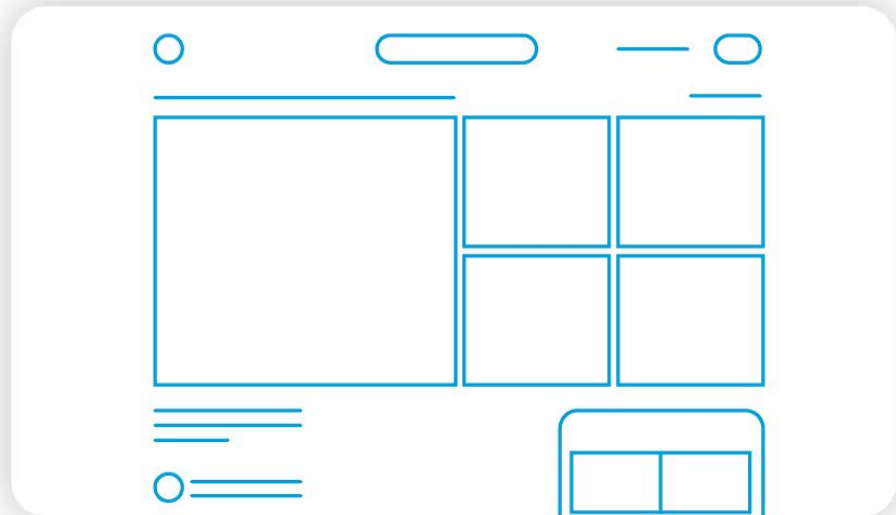
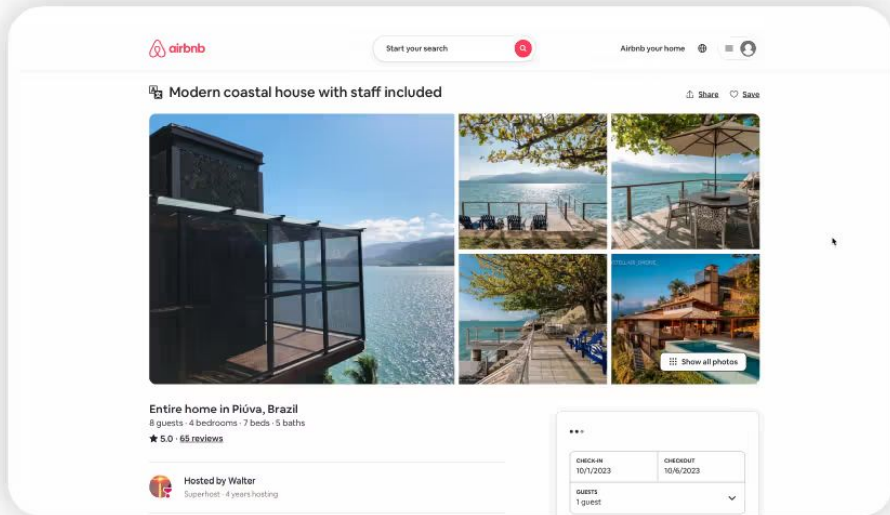




Pragnanz (German for “good figure”)

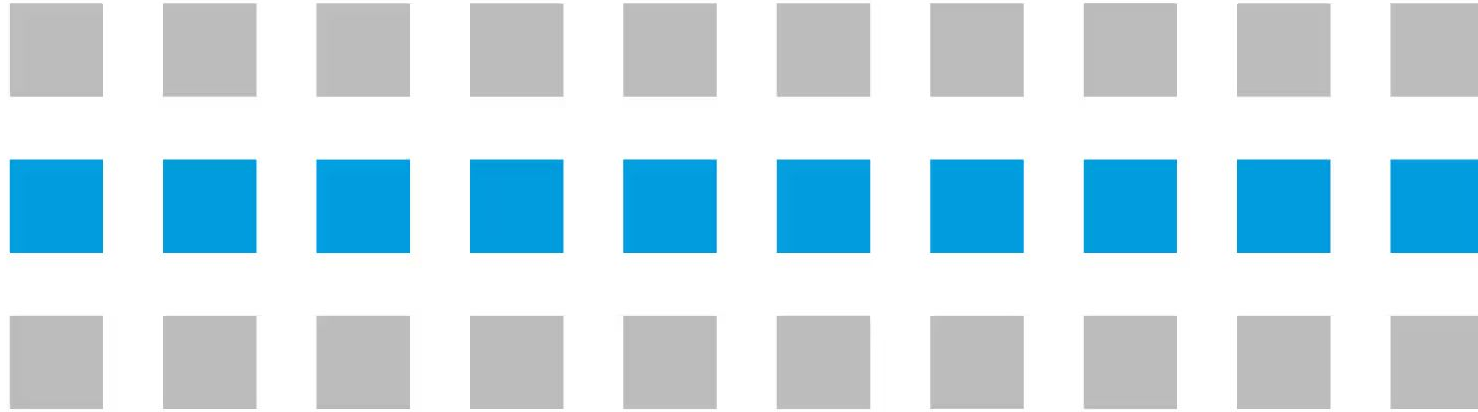
- We perceive/interpret visuals in the **simplest way possible**. Our brains automatically favor the **most stable, straightforward interpretation**.
 - Reflects our preference for symmetry, simple forms, & clear organization.
 - Guides designers to minimize visual complexity for easy understanding.
- E.g. Given a messy arrangement of coloured blocks on a table, your mind looks for basic shapes or clean lines first, rather than complex or “zigzag” patterns.

Pragnanz



Similarity

Gestalt Rule: **Similarity**





Similarity

- Elements sharing **similar characteristics** (colour, shape, size) are **mentally grouped together**.
 - Enables quick categorization based on shared features.
 - Helps you pick out familiar faces or patterns in a crowd—people wearing similar uniforms, for instance.
- A set of matching red chairs in a conference room stands out as a single group, separate from any green chairs, even if they're all arranged together in one space.



Apple Store Mac iPad iPhone Watch Vision AirPods TV & Home Entertainment Accessories Support

iPhone 16 Pro

Built for Apple Intelligence.

[Learn more](#) [Buy](#)

A side-profile view of an iPhone 16 Pro in a dark, metallic finish. The phone is positioned horizontally, with its triple-camera system and flash visible on the left. The background is dark, and the word 'PRO' is illuminated in large, glowing white letters behind the phone.

iPhone 16

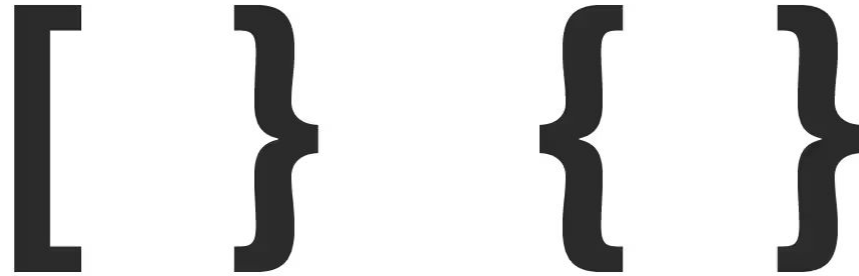
Built for Apple Intelligence.

[Learn more](#) [Buy](#)

A side-profile view of an iPhone 16 in a dark, metallic finish. The phone is positioned horizontally, with its single-camera system and flash visible on the left. The background is dark, and the word '16' is illuminated in large, glowing white letters behind the phone.

Symmetry and Order

Gestalt Rule: **Symmetry**





Symmetry and Order

- Symmetrical arrangements appear **more orderly & cohesive**. **Our brains love balance** & see symmetrical patterns as a unified whole.
 - Symmetry is often associated with beauty, harmony, and reliability.
 - Guides how we arrange items in a room or how we judge certain designs (e.g. infographics) to be aesthetically pleasing.
- A heart shape , mirrored from left to right, looks balanced and complete. You see it as a single symbol rather than two separate halves. 



Symmetry and Order

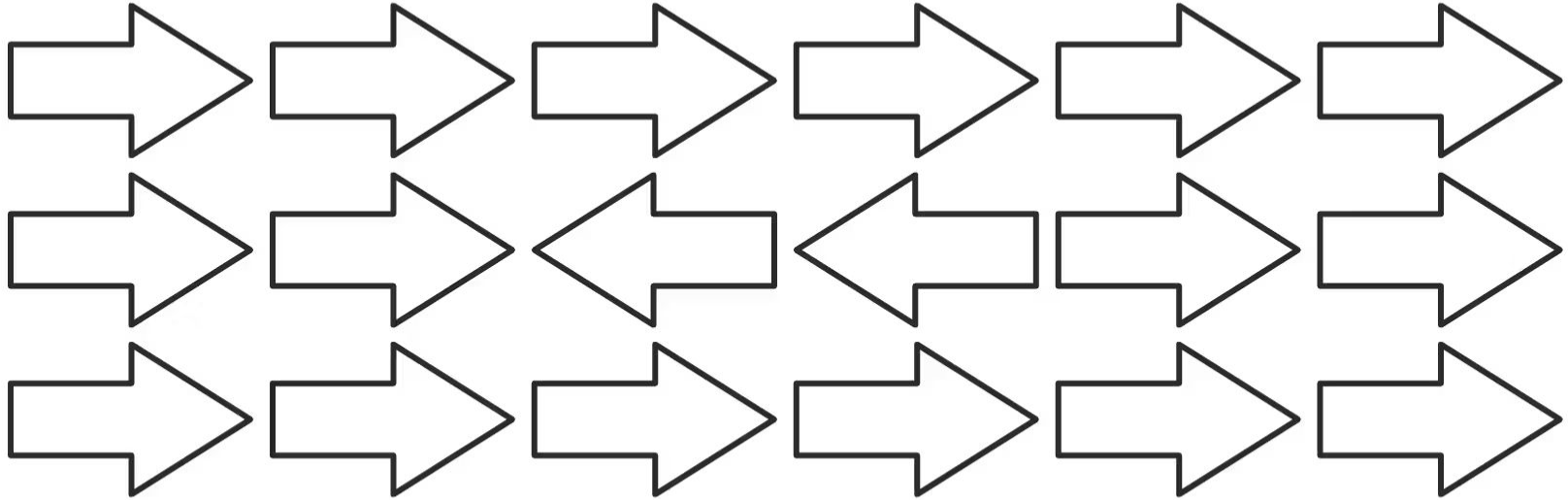
Google

Google Search

I'm Feeling Lucky

Common Fate

Gestalt Rule: **Common Fate**





Common Fate

- Elements moving in the **same direction** or showing the same change over time are perceived as **belonging together**.
 - Helps interpret motion and group dynamics in busy environments.
 - Applies to everything from following a parade to understanding traffic flows (people heading in the same direction are likely together).
- A flock of birds all heading south in formation is seen as a single group, even if they sometimes shift positions within that formation.



Common Fate

Questions related to Web Design

How to design a web page?



How much do web designers make?



How to become a web designer?



What does a web designer do?



Is web design coding?





Q: Which Gestalt Principles is most applicable to this image?



- A. Figure/Ground
- B. Invariance
- C. Pragnanz
- D. Similarity
- E. Symmetry and Order

following finding

Type the two words:

following finding

Send



Q: Which Gestalt Principles is most applicable to this image?



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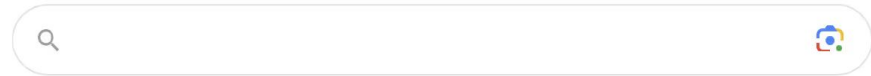
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Google Search

I'm Feeling Lucky



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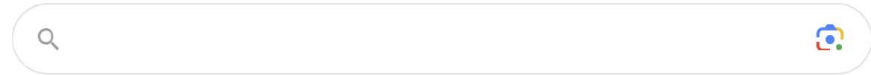
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B. Pragnanz

C. Closure

D. Symmetry and Order

E. Common Fate



Google Search

I'm Feeling Lucky



Learning Goals

After this **today's lecture**, you should be able to:

- Explain the **important of representation** as it relates to visualization design and usage.
- Assess the **suitability of a visualization** for a specific task.
- Compare **effective** and **flawed** visualizations using examples from media.
 - State the **strengths** and **weaknesses** of these data visualizations.
- Assess the **ethical responsibility of designers** in crafting visualizations that truthfully reflect data.

Implications of Visualizations

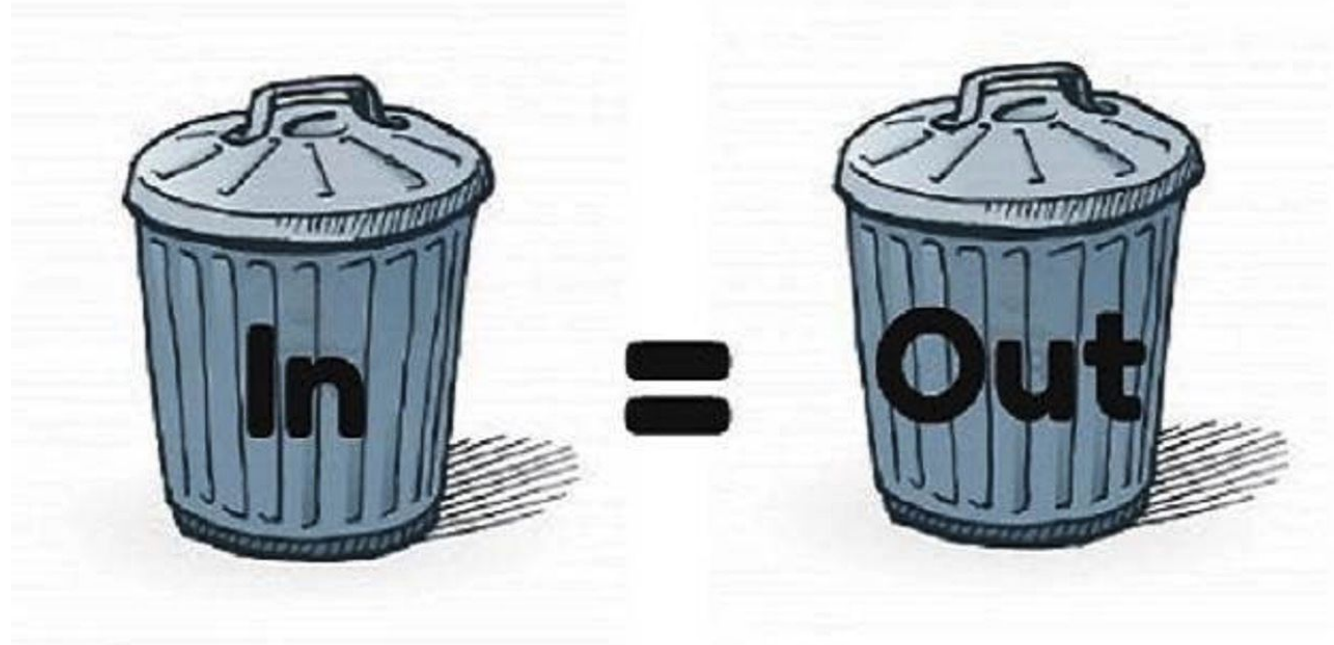


Ethical Considerations

- Visualizations are **NOT** facts
- A visualization's accuracy is dependent upon
 - The quality of the data used
 - The manner in which the data is conveyed
(i.e., its representation)
 - The objectivity of its creator (i.e., what's the goal/motive?)

Implications of Visualization

- Incorrect data
- Missing data
- Insufficient data





Implications of Visualization

- Truncated graphs
- Exaggerated scaling
- Incorrect Attribute to Channel Mapping
- Improper extraction
- Misusing metaphors
 - Is green good or bad? Is red good or bad?
 - Flawed representations

Examples & Discussion



AROUND
The
WORLD

GIANT HORNETS KILL 42 PEOPLE IN CHINA

LIVE
CNN

LIVE NOW

CHICAGO  71°

DETROIT  66°

HOUSTON  77°

HEALTH EVALUATION IN MIRIAM CAREY'S HOME AFTER DC C

CNN.com



<https://www.mediamatters.org/fox-news/dishonest-fox-charts-obamacare-enrollment-edition>

Dishonest Fox Charts: Obamacare Enrollment Edition

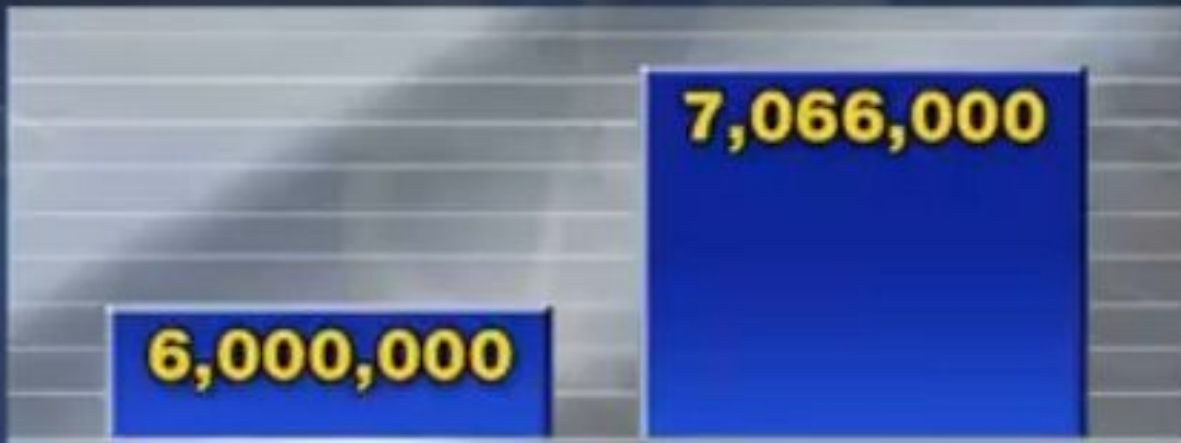
WRITTEN BY SAMANTHA WYATT

PUBLISHED 03/31/14 11:49 AM EDT

Fox News showed a dramatically skewed chart to suggest enrollment for the Affordable Care Act (ACA) would come up far short of the initial estimate of 7 million enrollees.

On March 27, health insurance enrollment through the ACA's exchanges [surpassed 6 million](#), exceeding the [revised estimate](#) of enrollees for the program's first year before the March 31 open enrollment deadline. Enrollment appears on track to hit the Congressional Budget Office's initial estimate of [7 million](#) sign-ups, and taking Medicaid enrollees into account, the ACA will have reportedly extended health care coverage to at least [9.5 million](#) previously uninsured individuals.

OBAMACARE ENROLLMENT



AS OF
MARCH 27

MARCH 31
GOAL



8:13 CT

SOURCE: HHS

STUDIO ESTIMATES... TEEN SCI-FI FLICK "DIVERGEN" S&P FUT ▲ 10.50

OBAMACARE ENROLLMENT



FOX NEWS
channel

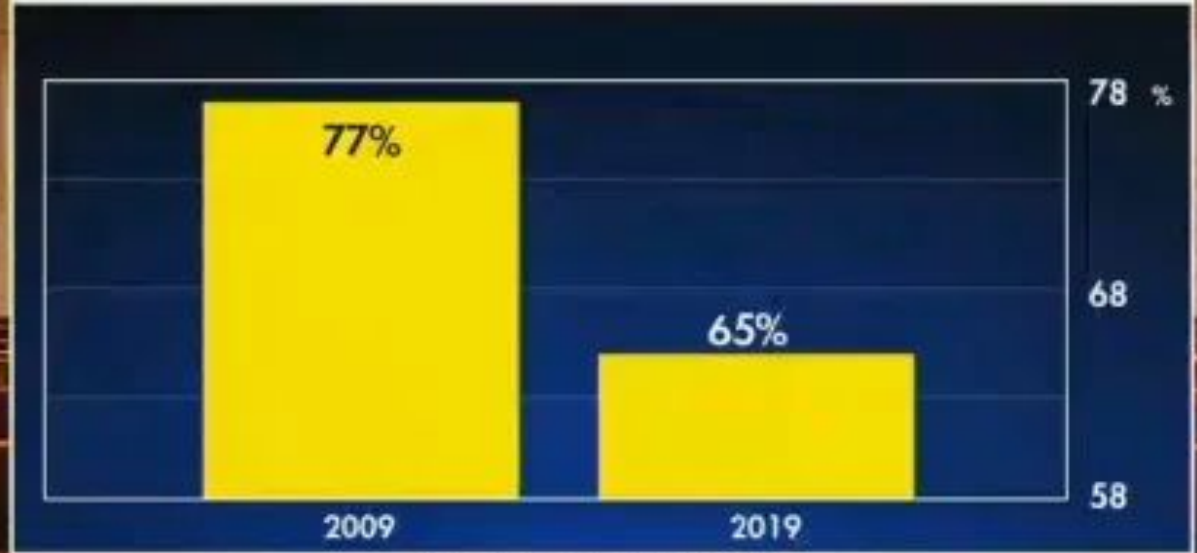
SOURCE: CBO

AMERICA'S
newsroom

RED DISCOUNTED FUEL TO FMR UKRAINIAN PRES V S&P FUT ▲ 6.50



PERCENTAGE OF AMERICANS THAT IDENTIFY AS CHRISTIANS



PEW RESEARCH CENTER

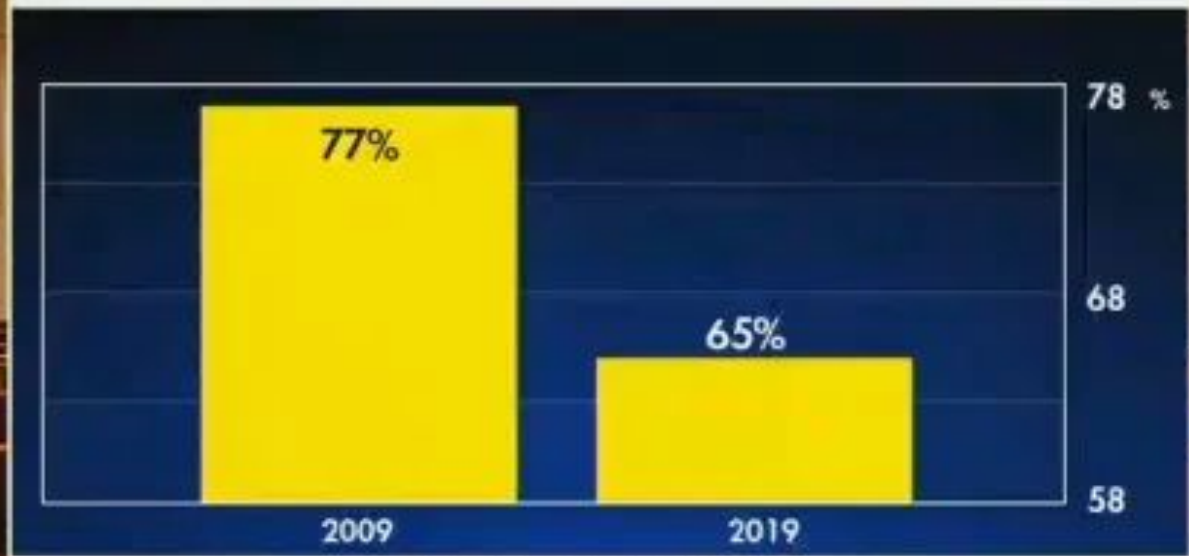


WE USED TO BE AN ENTHUSIASTICALLY CHRISTIAN NATION

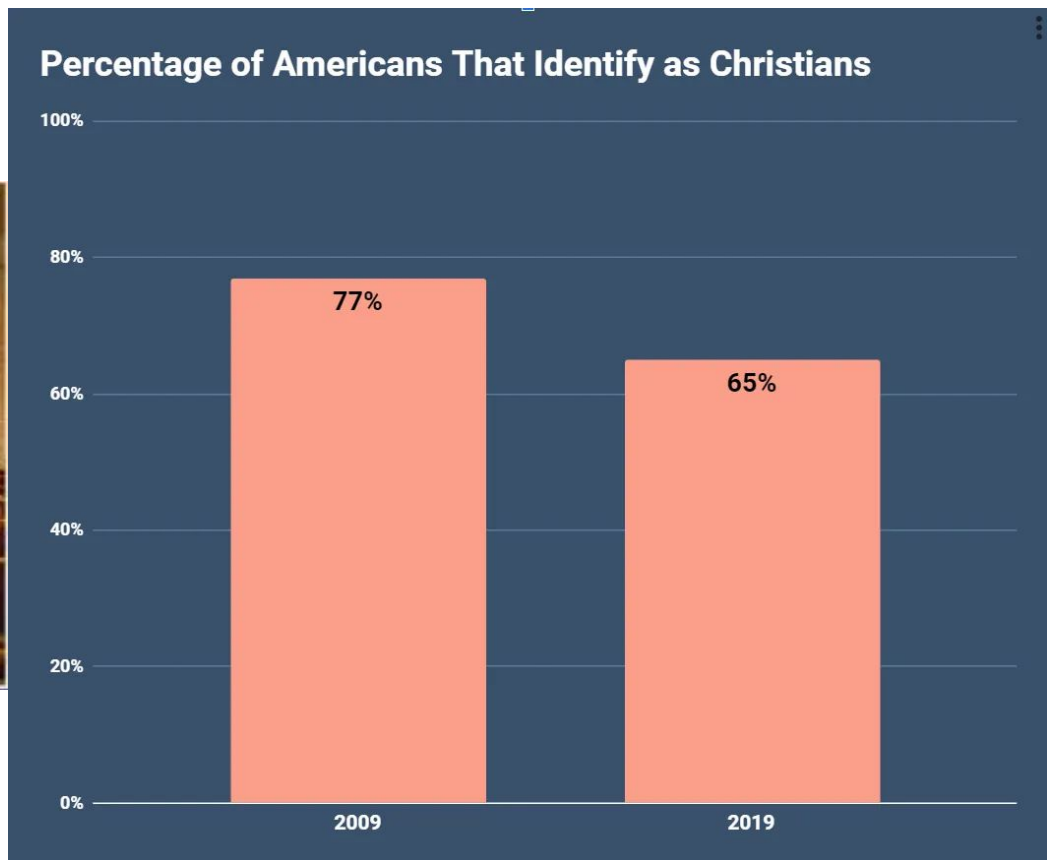
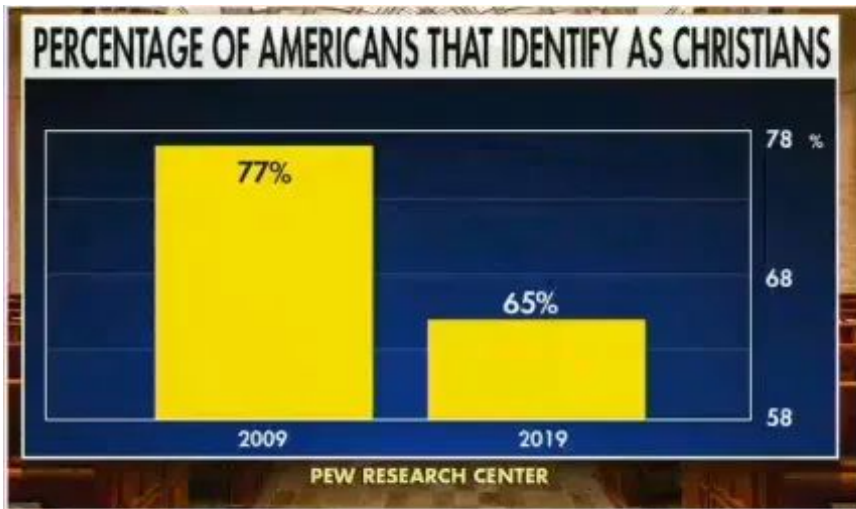
TUCKER CARLSON • TONIGHT •



PERCENTAGE OF AMERICANS THAT IDENTIFY AS CHRISTIANS



PEW RESEARCH CENTER



YANG WINS IOWA'S OFFICIAL YOUTH STRAW POLL!

YANG 22.5%

BERNIE 21%

BUTTIGIEG 17.9%

BIDEN 11.5%

WARREN 9.6%

SOURCE: IOWA SECRETARY OF STATE

Yang 2020

YANG WINS IOWA'S OFFICIAL YOUTH STRAW POLL!

YANG 22.5%

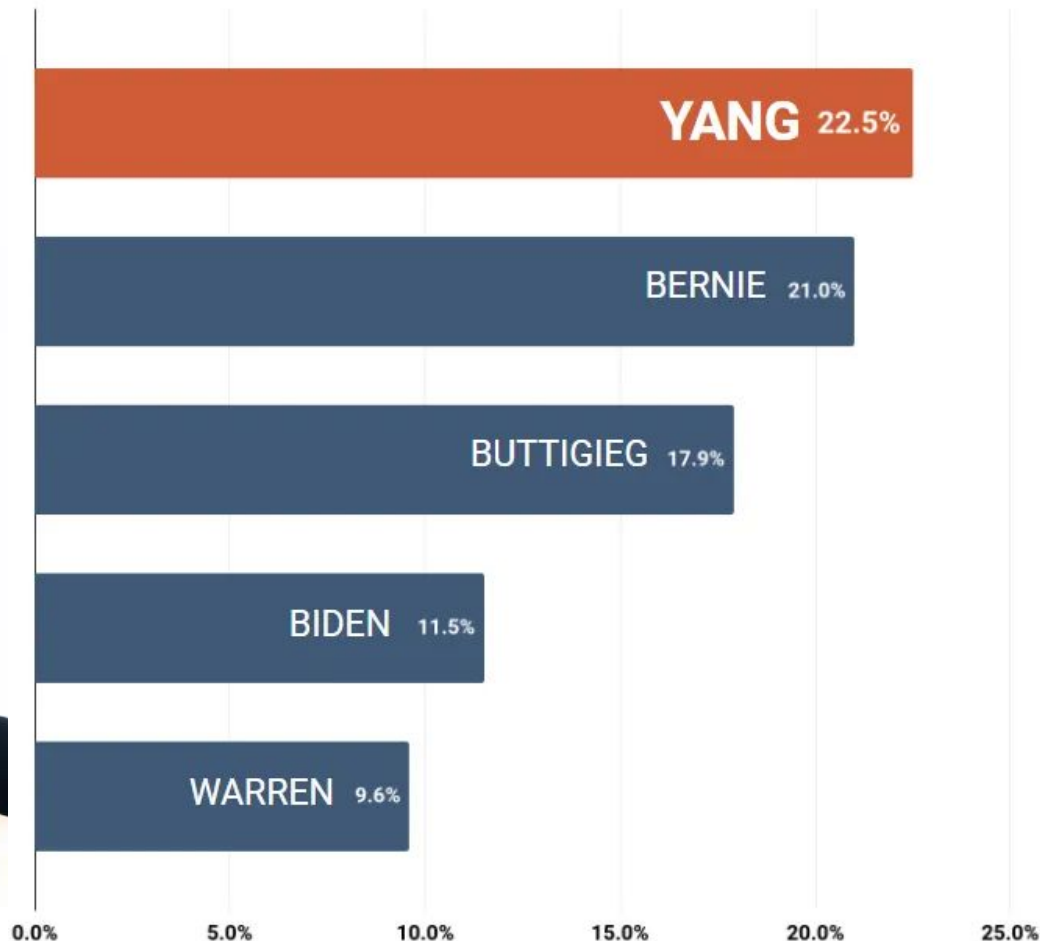
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BUTTIGIEG 17.9%

BIDEN 11.5%

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SOURCE: IOWA SECRETARY OF STATE





DAVIDsTEA

1d · 🌐

On a scale of 1 to Energizer Bunny, how much caffeine do you need today? Please refer to the chart above and plan accordingly 🧐

<https://bit.ly/DTShopAll>

caffeine ratings





DAVIDsTEA

1d • 🌍

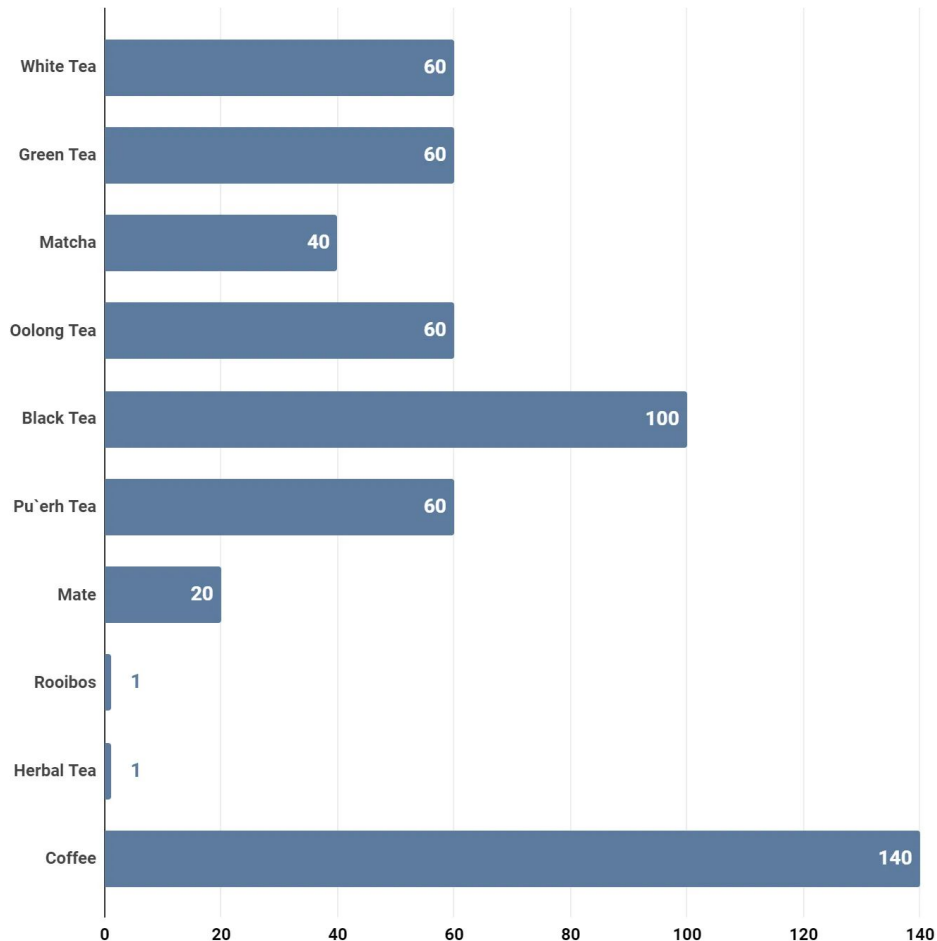
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caffeine ratings



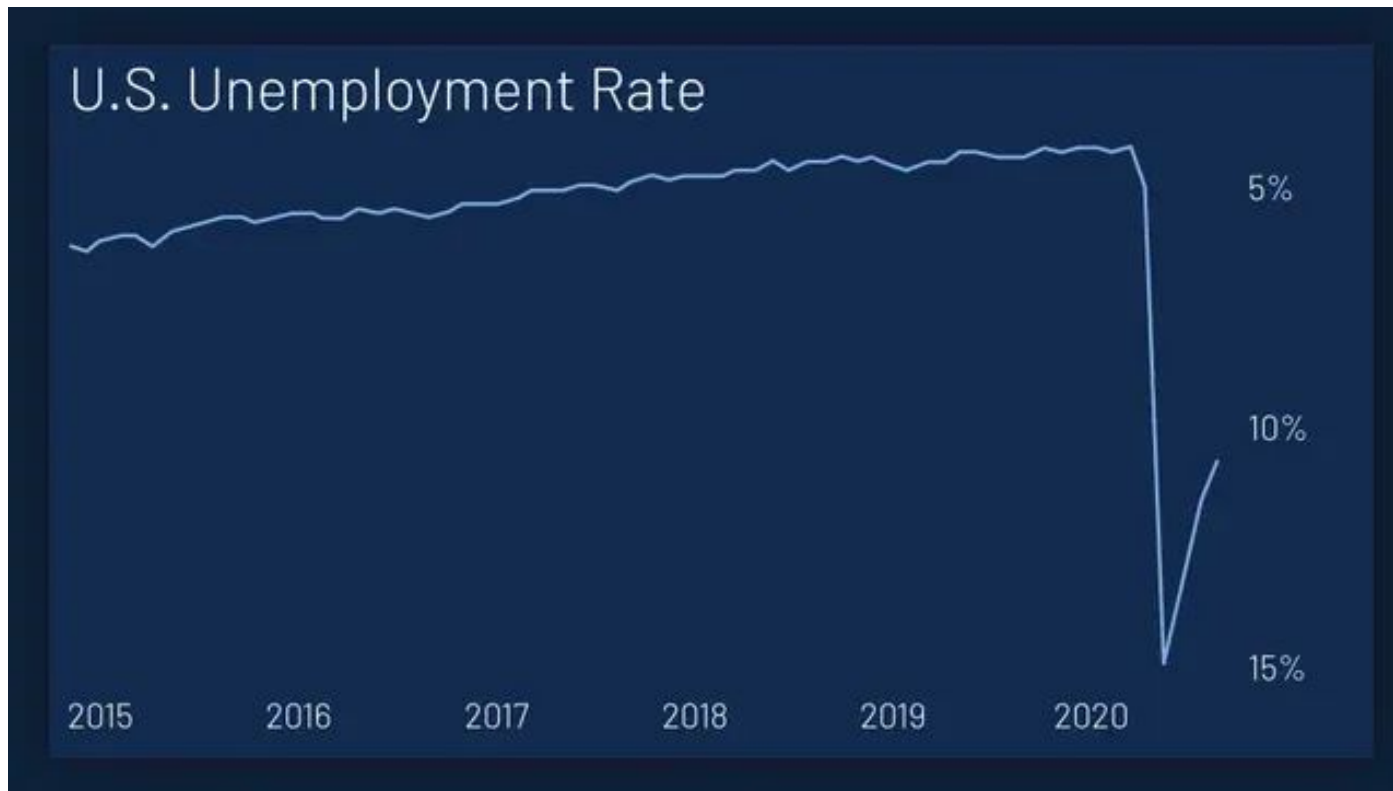
Caffeine Ratings



Q: What is wrong with the following graph?



iClicker



U.S. Unemployment Rate



U.S. Unemployment Rate



2015

2016

2017

2018

2019

2020

U.S. Unemployment Rate



U.S. Unemployment Rate



■ SE DESPLOMA EL COBRE POR EFECTO DEL CORONAVIRUS



FUENTE: AGENCIAS



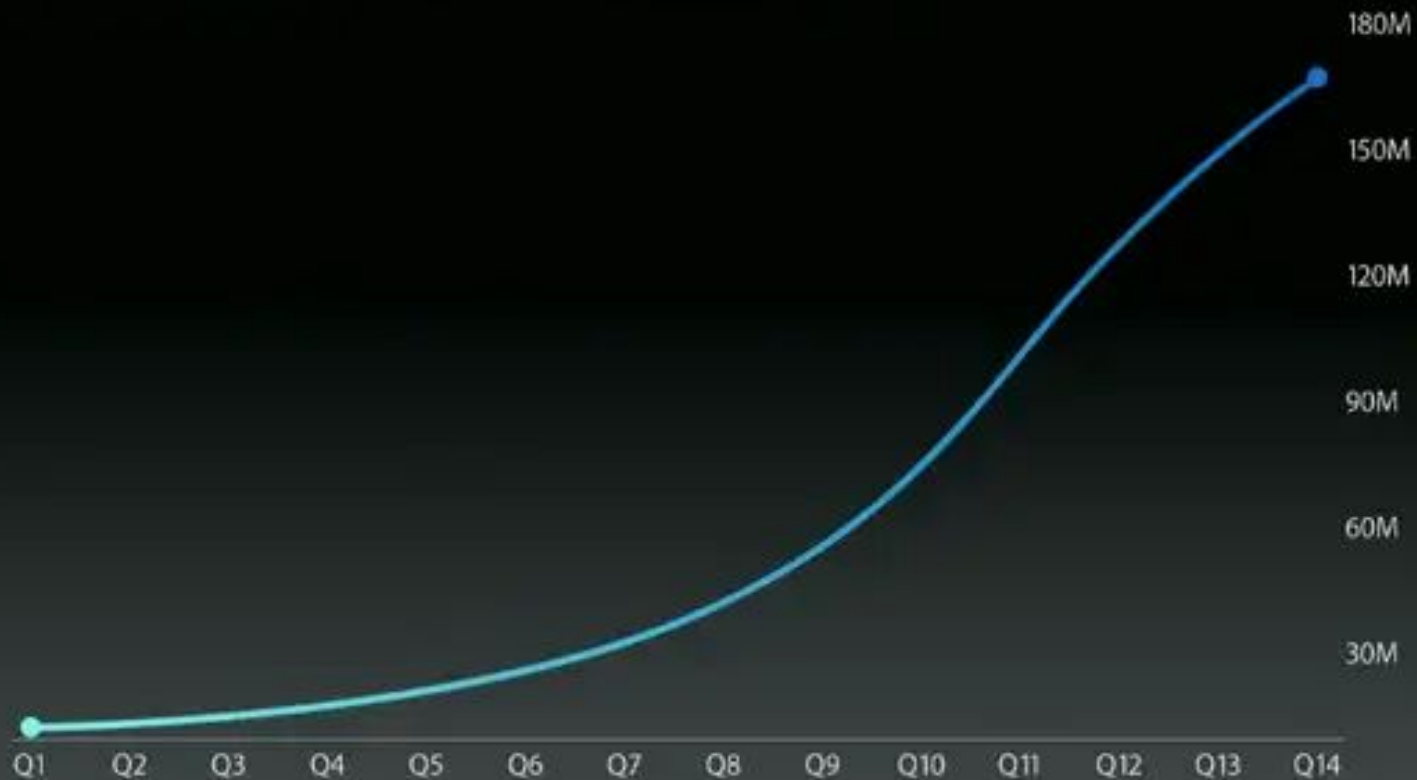
■ SE DESPLOMA EL COBRE POR EFECTO DEL CORONAVIRUS



FUENTE: AGENCIAS



Cumulative iPad sales



Cumulative iPad sales



M3 family performance cores

🍏 M3 family



🍏 M2 family



15% faster

🍏 M1 family



30% faster

COVID COMMUNITY LEVELS

AS OF JUNE 30

PROVIDENCE
LOW

KENT
MEDIUM

WASHINGTON
LOW

BRISTOL
LOW

NEWPORT
LOW



COVID COMMUNITY LEVELS

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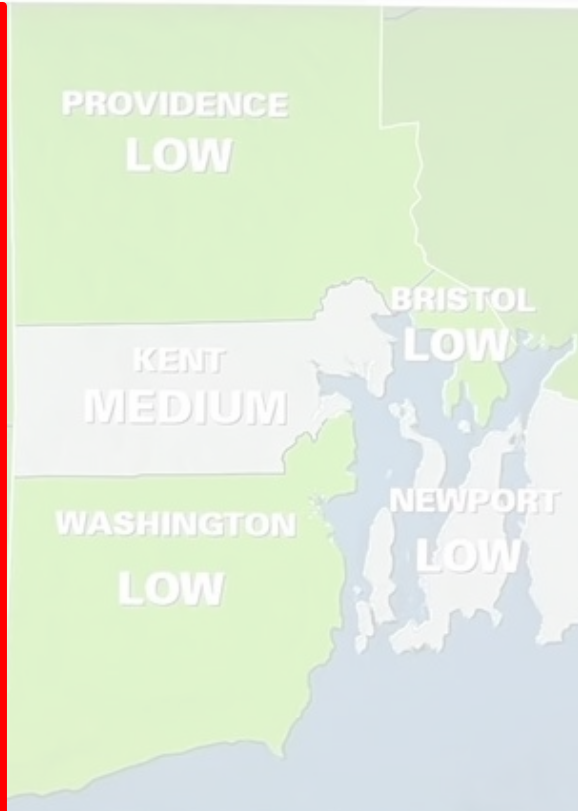
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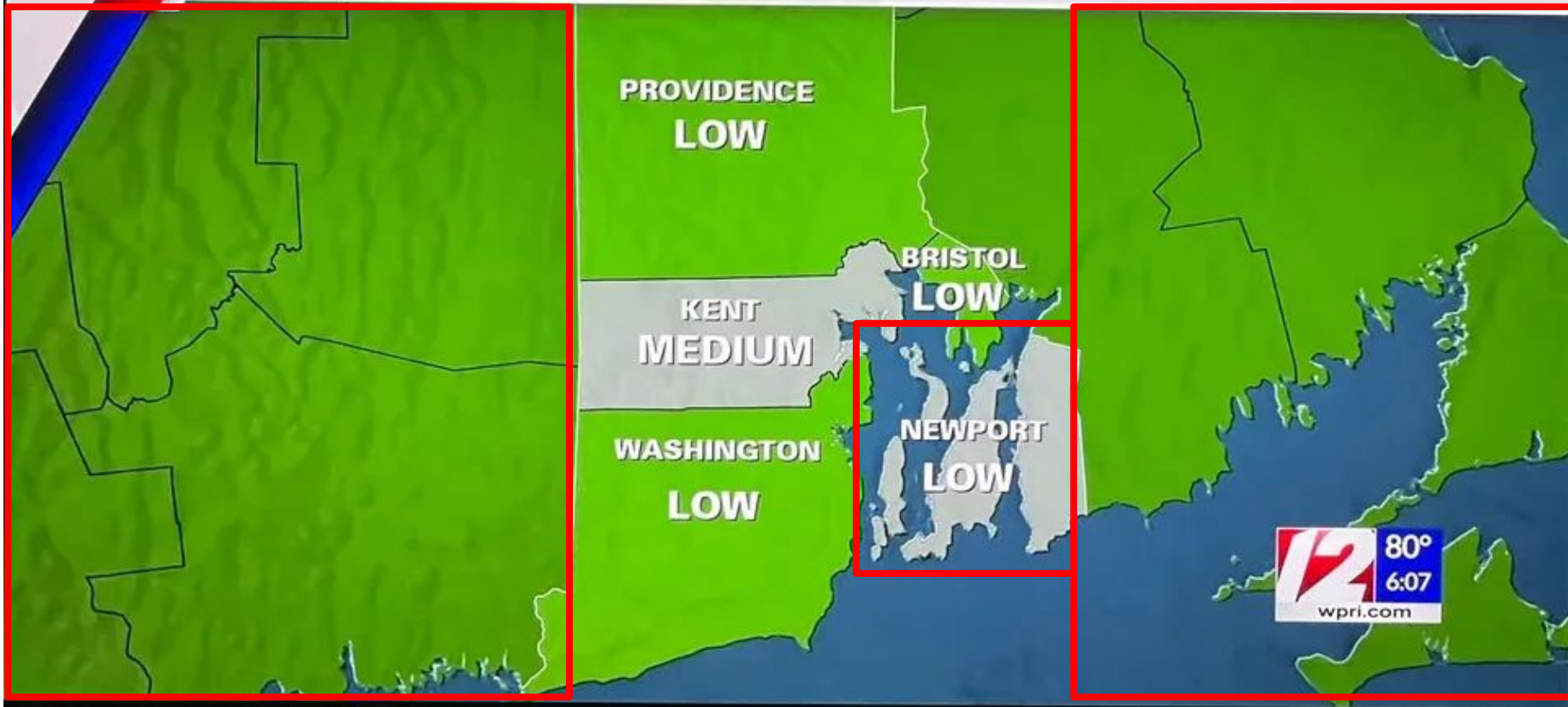
COVID COMMUNITY LEVELS

AS OF JUNE 30



COVID COMMUNITY LEVELS

AS OF JUNE 30



Take Home Reading

Visualization: Fact or Fiction?

<https://www.business.com/articles/data-visualization-downfalls/>

Fact or Fiction: Common Downfalls of Data Visualizations

Learn how to create better visuals in your presentations that will foster trust and communication.



Written by: Sanket Shah, Senior Editor

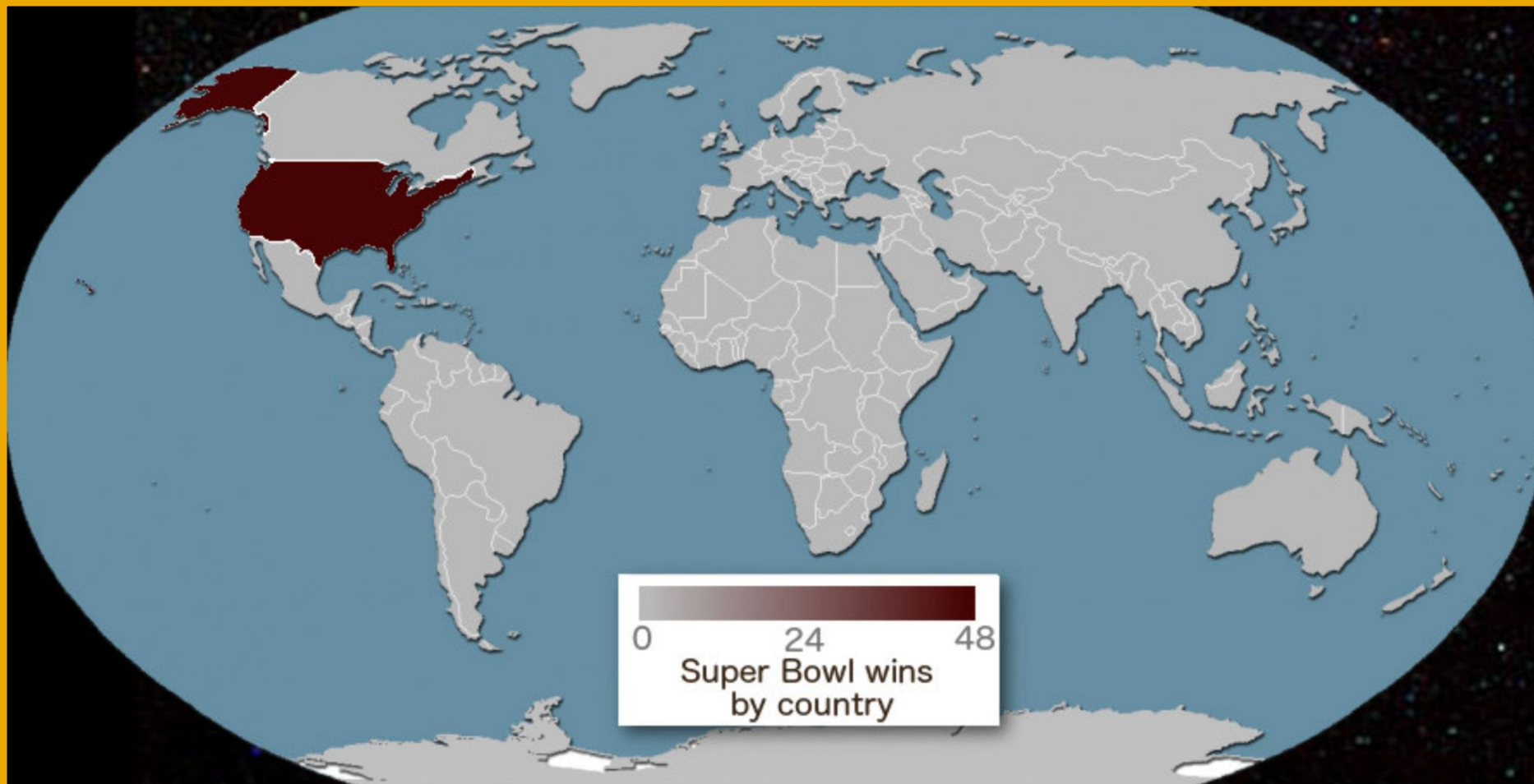


Editor Verified: Gretchen Grunburg, Senior Editor

Updated May 02, 2024

Take Home Activity

**What's wrong with
the following
visualizations?**



AMERICANS WHO HAVE TRIED MARIJUANA

CBS NEWS POLL

51%
TODAY

43%
LAST YEAR

34%
1997



Source: MOE +/- 4%

HIGH SUPPORT FOR LEGALIZING MARIJUANA

MORE THAN HALF OF AMERICANS SAY THEY'VE TRIED POT

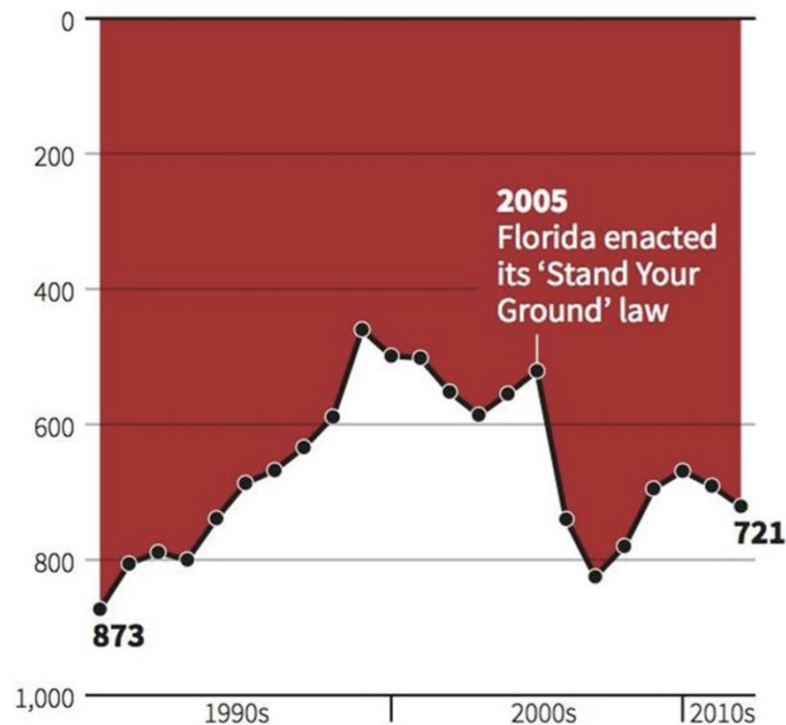


LIVE

CBSN

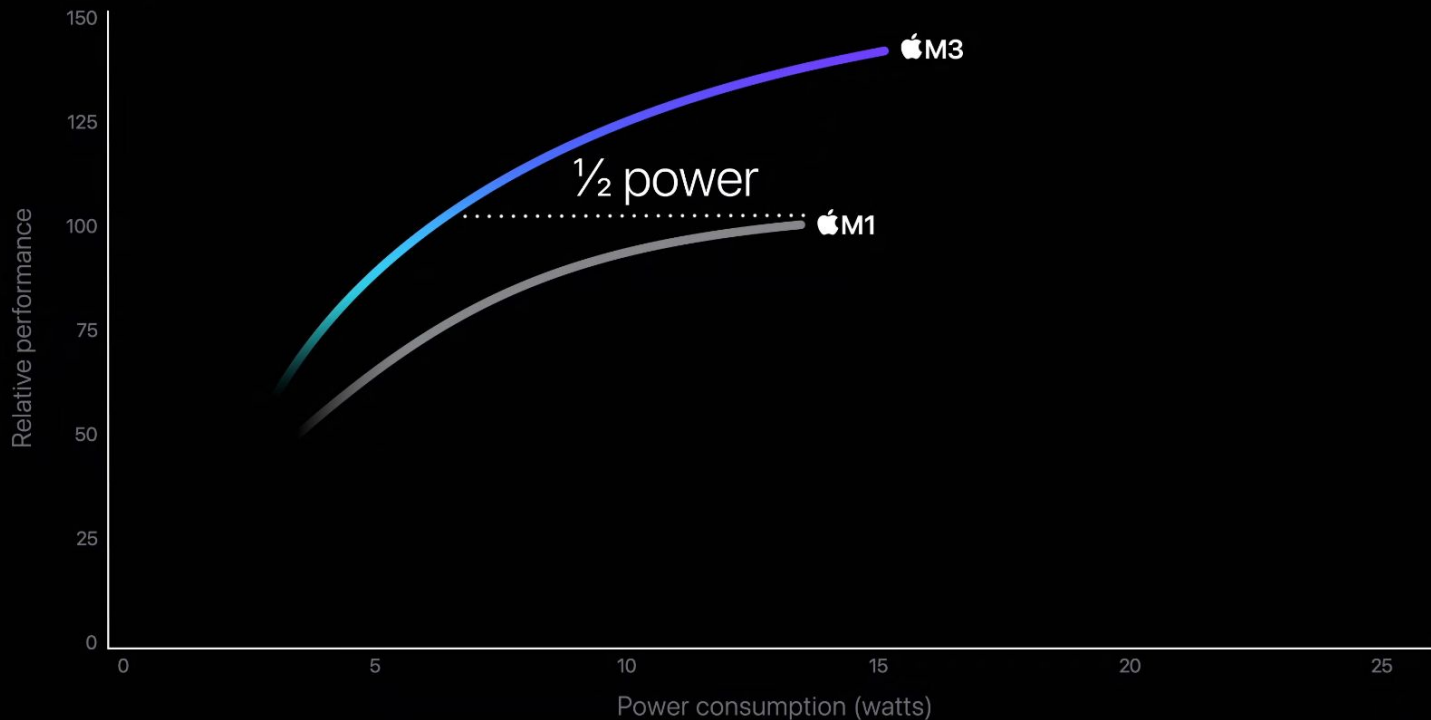
Gun deaths in Florida

Number of murders committed using firearms

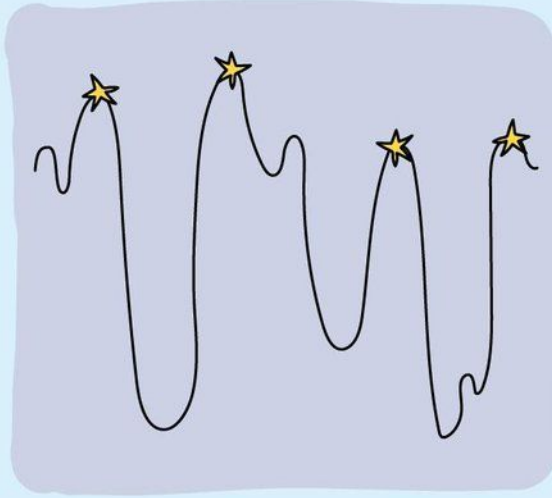


Source: Florida Department of Law Enforcement

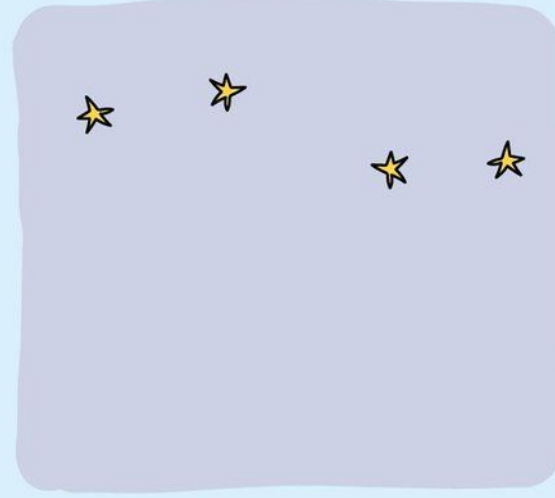
M3 CPU performance



LIFE



WHAT WE SEE
ON SOCIAL MEDIA



LIZ FOSSLIE